Project Idea 1

Title: City Bus Reservation System

Feature 1:Secure Login & Seat Map Visualization

Login verification using string comparison loops. Once authenticated, the system renders the current bus layout by traversing an array, distinguishing between 'Empty' (0) and 'Booked' (1) seats in a grid format.

Feature 2: Variable Ticket Booking & Validation

Allows users to input specific seat numbers (supporting variable-length inputs like "5" or "12"). The system validates the input against the array index to ensure the seat is not already taken before updating the status to 'Booked'.

Feature 3: Dynamic Fare Calculation & Categorization

Calculates the total fare based on passenger category (Student vs. Regular). The system uses conditional logic to apply discounts and multiplication algorithms to handle ticket quantities, updating the session's total payable amount.

Feature 4: Admin: Revenue Dashboard

A password-protected Admin view that aggregates daily sales data. It uses a loop to traverse the sales variables, calculating the total number of seats sold and the total revenue generated, providing a "End of Day" summary.

Feature 5:Ticket Cancellation & Refund Logic

Allows a user to cancel a previously booked ticket. The system verifies the seat was actually booked, resets the specific array index to '0' (Empty), and performs subtraction logic to deduct the fare from the total revenue variable.

Feature 6:Search Passenger & Seat Status

A lookup algorithm where the user inputs a Seat ID. The system directly accesses the array index to return the specific status (Available/Booked) and price category, allowing quick checks without reloading the full map.

Project Idea 2

Title: University Cafe Management System

Feature 1:Auth System & Live Menu Display

Features a dual-login system: an 8-digit ID check for Students and a passcode for Admins. Upon login, the system retrieves item data from memory arrays (Prices and Stock) to display a dynamic menu that hides out-of-stock items.

Feature 2:Multi-Item Ordering System

Allows students to select Item IDs and input variable quantities (e.g., "3 Burgers"). The system performs a logic check (Order_Qty <= Stock_Qty); if valid, it decrements the inventory array and adds the cost to the user's temporary bill.

Feature 3:Bill Generation with Tax Logic

A checkout feature that takes the accumulated session total and calculates the final bill. It includes a sub-routine to calculate 5% VAT on the total amount using multiplication/division registers and displays the net payable cash.

Feature 4:Admin: Inventory Restock

A specialized view for Admins to manually update the Stock Array. It allows adding variable amounts to existing stock levels (e.g., inputting "50" adds 50 units to the current count), preventing "Sold Out" errors.

Feature 5:Admin: Price Adjustment

Allows the Admin to select an item and overwrite its value in the Price Array. This ensures the system supports dynamic data changes rather than static constants, meeting the requirement for complex memory handling.

Feature 6: Sales Reporting & Analysis

A reporting feature that tracks the total count of items sold per category during a session. It iterates through the sales array to identify and display which item was the "Best Seller" based on quantity sold.